Experiment 11: Orchestration using Docker Compose & Docker Swarm

---

Objective

To understand container orchestration using Docker Swarm by deploying, scaling, and managing a multi-container WordPress application.

Theory

What is Docker Swarm?

Docker Swarm is a container orchestration tool that allows you to manage a cluster of Docker nodes as a single system. It provides features like:

- Service management
- Load balancing
- Scaling
- Self-healing

Key Concepts

Term	Description
Node	A machine in the Swarm cluster
Manager	Controls the cluster
Worker	Executes tasks
Service	Definition of container(s)
Stack	Group of services
Replica	Number of container instances

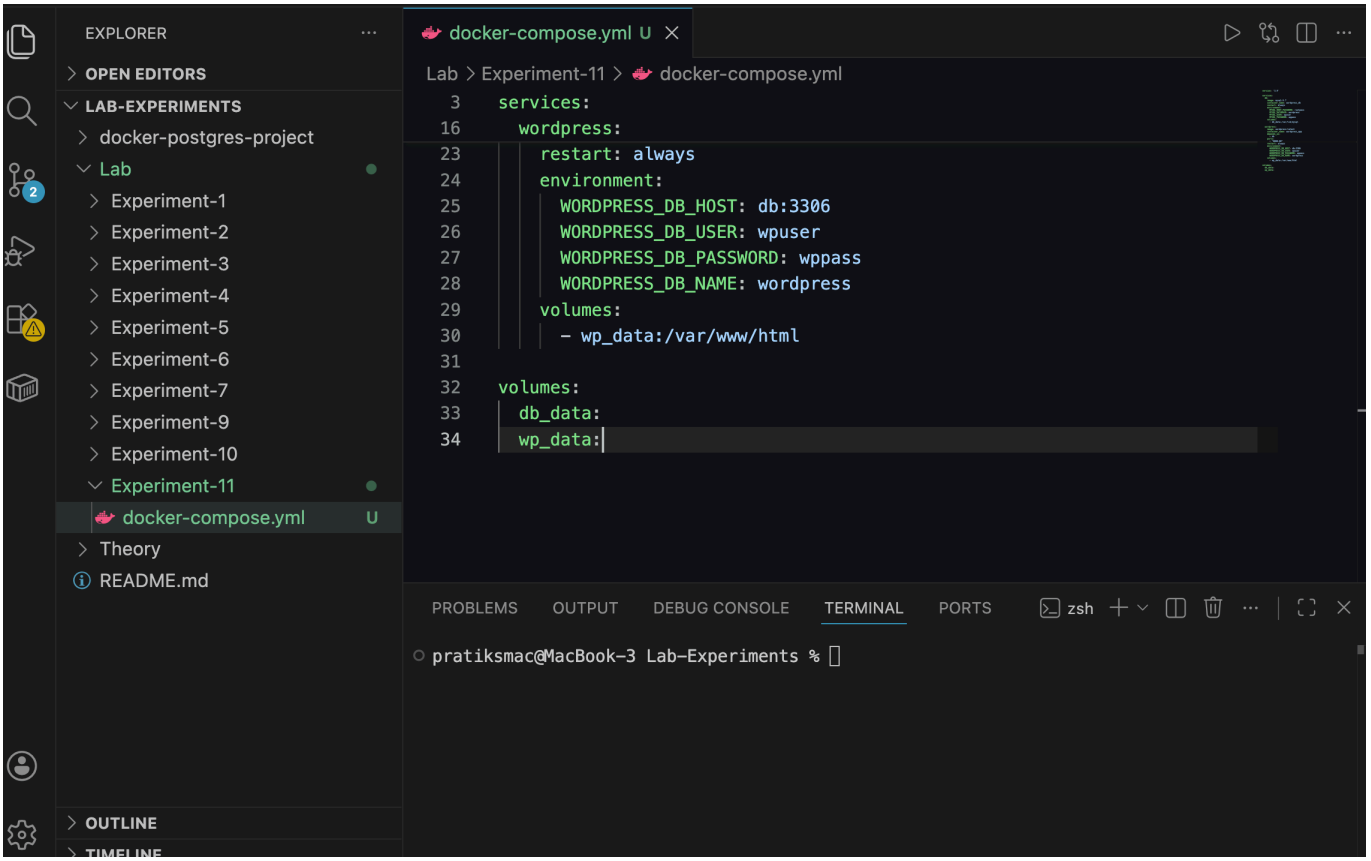
Tools Used

- Docker Desktop
- Docker Compose
- Docker Swarm

Implementation Steps

Step 1: Create Project Directory

```
mkdir Experiment-11
cd Experiment-11
```



Step 2: Create docker-compose.yml

```yaml
version: '3.9'

services:
  db:
    image: mariadb:10.6
    environment:
      MYSQL_ROOT_PASSWORD: rootpass
      MYSQL_DATABASE: wordpress
      MYSQL_USER: wpuser
      MYSQL_PASSWORD: wppass
    volumes:
      - db_data:/var/lib/mysql

  wordpress:
    image: wordpress:latest
    ports:
      - "8081:80"
    environment:
      WORDPRESS_DB_HOST: db:3306
```

```
      WORDPRESS_DB_USER: wpuser
      WORDPRESS_DB_PASSWORD: wppass
      WORDPRESS_DB_NAME: wordpress
    volumes:
      - wp_data:/var/www/html
    deploy:
      restart_policy:
        condition: on-failure

volumes:
  db_data:
  wp_data:
```

Step 3: Clean Previous Setup

```
docker compose down -v
docker rm -f $(docker ps -aq)
docker network prune -f
```

Step 4: Initialize Docker Swarm

```
docker swarm init
```

Verify:

```
docker node ls
```

```
pratiksmac@MacBook-3 ~ % docker node ls
ID                            HOSTNAME         STATUS    AVAILABILITY   MANAGER STATUS   ENGINE VERSION
imy3eyxr7vqxm0a8nw0viq1dt *   docker-desktop   Ready     Active         Leader           29.2.0
```

Step 5: Deploy Stack

```
docker stack deploy -c docker-compose.yml wpstack
```

```
pratiksmac@MacBook-3 Experiment-11 % docker stack deploy -c docker-compose.yml wpstack
Ignoring unsupported options: restart

Ignoring deprecated options:

container_name: Setting the container name is not supported.

Since --detach=false was not specified, tasks will be created in the background.
In a future release, --detach=false will become the default.
Creating network wpstack_default
Creating service wpstack_db
Creating service wpstack_wordpress
pratiksmac@MacBook-3 Experiment-11 %
```

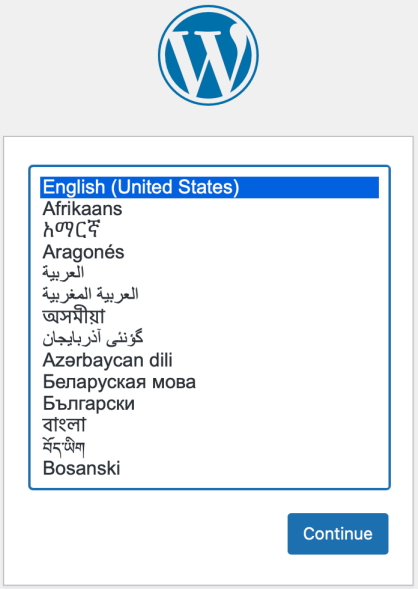
Step 6: Verify Services

```
docker service ls
```

```
pratiksmac@MacBook-3 Experiment-11 % docker service ls
ID             NAME                MODE         REPLICAS   IMAGE              PORTS
7cqu6aei5iio   wpstack_db          replicated   0/1        mysql:5.7
e345b6gxfq7h   wpstack_wordpress   replicated   1/1        wordpress:latest   *:8080->80/tcp
pratiksmac@MacBook-3 Experiment-11 %
```

Step 7: Check Running Containers

```
docker ps
```



Step 8: Access Application

Open browser:

```
http://localhost:8081
```

```
pratiksmac@MacBook-3 Experiment-11 % docker service scale wpstack_wordpress=3
wpstack_wordpress scaled to 3
overall progress: 3 out of 3 tasks
1/3: running   [==================================================>]
2/3: running   [==================================================>]
3/3: running   [==================================================>]
verify: Service wpstack_wordpress converged
pratiksmac@MacBook-3 Experiment-11 %
```

```
pratiksmac@MacBook-3 Experiment-11 % docker service ls
ID             NAME                MODE         REPLICAS   IMAGE              PORTS
7cqu6aei5iio   wpstack_db          replicated   1/1        mysql:8.0
e345b6gxfq7h   wpstack_wordpress   replicated   3/3        wordpress:latest   *:8080->80/tcp
pratiksmac@MacBook-3 Experiment-11 %
```

Step 9: Scale Application

```
docker service scale wpstack_wordpress=3
```

Verify:

```
docker service ls
```

```
pratiksmac@MacBook-3 Experiment-11 % docker service ps wpstack_wordpress
ID             NAME                IMAGE              NODE             DESIRED STATE   CURRENT STATE               ERROR     PORTS
orpsqs4q3pb2   wpstack_wordpress.1 wordpress:latest   docker-desktop   Running         Running 47 minutes ago
npy7dt0dpzuv   wpstack_wordpress.2 wordpress:latest   docker-desktop   Running         Running about a minute ago
1z32xansyeeh   wpstack_wordpress.3 wordpress:latest   docker-desktop   Running         Running about a minute ago
pratiksmac@MacBook-3 Experiment-11 %
```

```
pratiksmac@MacBook-3 Experiment-11 % docker ps | grep wordpress
d7b547ae1d59   wordpress:latest   "docker-entrypoint.s…"   3 minutes ago    Up 3 minutes    80/tcp    wpstack_wordpress.3.1z32xansyeehw29o4tpr8g814
2d4c9527473d   wordpress:latest   "docker-entrypoint.s…"   3 minutes ago    Up 3 minutes    80/tcp    wpstack_wordpress.2.npy7dt0dpzuvy6afxj7hne5uf
60705fc6845c   wordpress:latest   "docker-entrypoint.s…"   49 minutes ago   Up 49 minutes   80/tcp    wpstack_wordpress.1.orpsqs4q3pb2fx0kc1z8yjt0f
pratiksmac@MacBook-3 Experiment-11 %
```

Step 10: Test Self-Healing

```
docker ps | grep wordpress
docker kill <container-id>
docker service ps wpstack_wordpress
```

```
[pratiksmac@MacBook-3 Experiment-11 % docker kill d7b547ae1d59
d7b547ae1d59
[pratiksmac@MacBook-3 Experiment-11 % docker service ps wpstack_wordpress
ID             NAME                      IMAGE              NODE             DESIRED STATE   CURRENT STATE           ERROR                       PORTS
orpsqs4q3pb2   wpstack_wordpress.1       wordpress:latest   docker-desktop   Running         Running 56 minutes ago
npy7dt0dpzuv   wpstack_wordpress.2       wordpress:latest   docker-desktop   Running         Running 9 minutes ago
cub0i4eb8ikk   wpstack_wordpress.3       wordpress:latest   docker-desktop   Running         Running 8 seconds ago
1z32xansyeeh    \_ wpstack_wordpress.3   wordpress:latest   docker-desktop   Shutdown        Failed 13 seconds ago   "task: non-zero exit (137)"
pratiksmac@MacBook-3 Experiment-11 %
```

```
[pratiksmac@MacBook-3 Experiment-11 % docker stack rm wpstack
Removing service wpstack_db
Removing service wpstack_wordpress
Removing network wpstack_default
[pratiksmac@MacBook-3 Experiment-11 % docker service ls
ID        NAME      MODE      REPLICAS   IMAGE     PORTS
[pratiksmac@MacBook-3 Experiment-11 % docker ps
CONTAINER ID   IMAGE     COMMAND   CREATED   STATUS    PORTS     NAMES
pratiksmac@MacBook-3 Experiment-11 %
```

Step 11: Remove Stack

```
docker stack rm wpstack
```

Result

- Docker Swarm initialized successfully
- WordPress stack deployed using Docker Compose
- Services scaled to multiple replicas
- Self-healing demonstrated successfully
- Stack removed after completion

Conclusion

Docker Swarm enables efficient orchestration of containers by providing features like scaling, load balancing, and automatic recovery, making it suitable for managing distributed applications.

Key Takeaways

- Swarm converts Docker into a cluster manager
- Services manage containers automatically
- Scaling increases availability
- Self-healing ensures reliability
- Stack simplifies multi-container deployment